

**ECON 101—Introduction to Microeconomics
Fall 2023**

Midterm II

October 28, 2023

Name:

NU ID number:

Instructions:

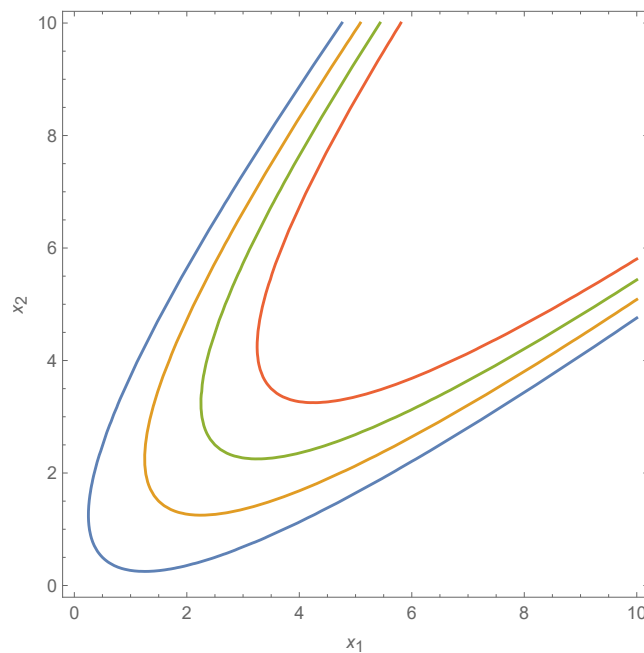
- This is a multiple choice exam, with 100 total points. Each question is worth 5 points. You will be provided scratch paper for your preliminary calculations but your official answers should be written **in the answer boxes below**. Answers written in scratch paper or outside the space provided will not be considered!
- The exam has a duration of 2 hours, from 16:00 to 18:00.
- You can request a bathroom break, but only one person can be out at any given time.
- On your desk, you are allowed to have your ID, writing instruments, a ruler, and a simple calculator (no graphing calculators). You may also have a simple watch if needed (smartwatches are not allowed).
- Please place all other personal items, including smart devices, to the side of the room.
- If you complete the exam early, you may leave the room, but not before 17:45. Otherwise, please remain in the room quietly until the exam concludes.

Multiple choice. Please write your answers here!

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Do not write below this area!

Ali has preferences described by the utility function $u(x_1, x_2) = 2(x_1 + x_2) - (x_1 - x_2)^2$. His indifference curve map, for four different levels of utility, is represented in the graphic below. The utility is increasing in the northeast direction, i.e., a higher indifference curve corresponds to a higher level of utility.



Furthermore, the marginal utility of good 1 is $MU_1 = 2 - 2(x_1 - x_2)$ and the marginal utility of good 2 is $MU_2 = 2 + 2(x_1 - x_2)$. Answer the following questions:

1. The utility function u above represents consumer's preferences that are:

- (a) Complete, Monotone, and Convex, but not Transitive.
- (b) Complete and Transitive, and Monotone, but not Convex.
- (c) Complete, Transitive and Convex, but not Monotone.
- (d) Complete, Transitive, Monotone, and Convex.
- (e) Transitive, Monotone, and Convex, but not Complete.
- (f) None of the above.

2. The Marginal Rate of substitution of good 1 for good 2 is given by

$$\begin{array}{llll} \frac{x_1 - x_2}{x_1 + x_2} & \frac{x_1 + x_2}{x_1 - x_2} & \frac{1 - (x_1 - x_2)}{1 + (x_1 - x_2)} & \frac{2 + 2(x_1 - x_2)}{2 - 2(x_1 - x_2)} \\ \text{(a)} & \text{(b)} & \text{(c)} & \text{(d)} \end{array}$$

3. If the price ratio is $p_1/p_2 = 1$, the equation for the *income consumption curve*, in terms of x_1 and x_2 is given by

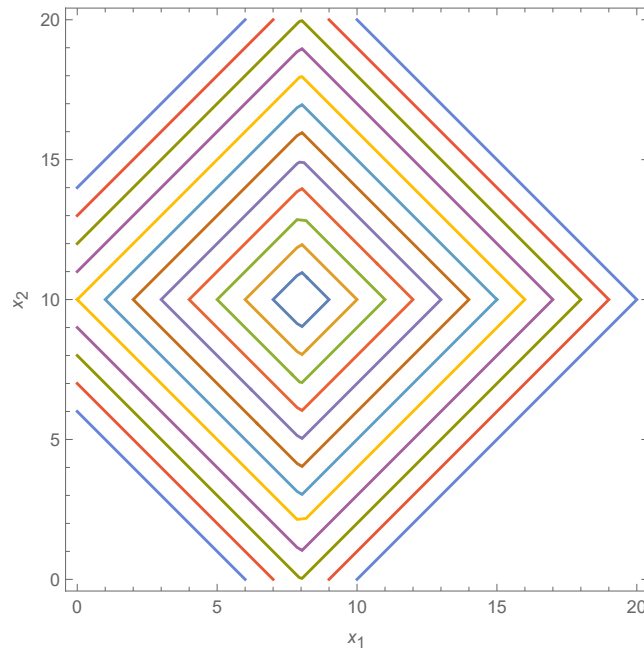
- (a) $x_2 = 2x_1$
- (b) $x_2 = x_1$
- (c) $x_2 = 2 + x_1$
- (d) $x_2 = \frac{1}{2}x_1$

4. Suppose that income is $I = 50$ and $p_1 = p_2 = 10$. Then the consumer's optimal consumption bundle (x_1^*, x_2^*) is:

- (a) $(5/2, 5/2)$
- (b) $(5/3, 10/3)$
- (c) $(3/2, 7/2)$
- (d) $(10/3, 5/3)$

5. If prices are $p_1 = 60$ and $p_2 = 20$, how large income I must be in order to get a strictly positive demand for good 1?
- (a) $I > 10$.
 - (b) $I > 20$.
 - (c) $I > 25$.
 - (d) $I > 80$.

Abuzer has preferences described by the utility function $u(x_1, x_2) = 10 - |x_1 - 8| - |x_2 - 10|$. Below you can find his indifference curve map for different levels of utility. Notice that utility increases towards the bundle $(8, 10)$.



6. Which of the following utility functions also represents Abuzer's preferences?
- (a) $\exp(-|x_1 - 8| - |x_2 - 10|)$
 - (b) $8 - |x_1 - 8| - |x_2 - 10|$
 - (c) $(10 - |x_1 - 8| - |x_2 - 10|)^3$
 - (d) $-|x_1 - 8| - |x_2 - 10|$
 - (e) All of the above
7. Suppose that income is $I = 100$ and prices are $p_1 = 3$ and $p_2 = 7$. The optimal consumption bundle for Abuzer is?
- (a) $(8, 10)$
 - (b) $(3, 13)$
 - (c) $(10, 10)$
 - (d) $(10, 8)$
 - (e) All of the above.

8. Suppose that prices are the same, $p_1 = 3$ and $p_2 = 7$, but income is $I = 90$. The optimal consumption bundle for Abuzer is?

- (a) (2, 12)
- (b) (8, 81/7)
- (c) (20/3, 10)
- (d) (10, 8)
- (e) (8, 10)
- (f) (9, 9)

9. Suppose that the price ratio is $p_1/p_2 < 1$. Then Abuzer's demand system, (x_1^m, x_2^m) is:

- (a) $(8, \frac{I}{p_2} - \frac{p_1}{p_2} 8)$
- (b) $(\frac{I}{p_2} - \frac{p_1}{p_2} 8, 8)$
- (c) $(\frac{I}{p_1} - \frac{p_2}{p_1} 10, 10)$
- (d) $(8, \frac{I}{p_1} - \frac{p_2}{p_1} 10)$

10. Suppose that the price ratio is $p_1/p_2 > 1$. Then Abuzer's demand system, (x_1^m, x_2^m) is:

- (a) $(8, \frac{I}{p_1} - \frac{p_2}{p_1} 10)$
- (b) $(\frac{I}{p_2} - \frac{p_1}{p_2} 8, 8)$
- (c) $(\frac{I}{p_1} - \frac{p_2}{p_1} 10, 10)$
- (d) $(8, \frac{I}{p_2} - \frac{p_1}{p_2} 8)$

Shynar has a utility function over a consumption good c , and money m given by $u(c, m) = m + \ln(c)$, where $\ln$ denotes the natural logarithm. The marginal utility for money is given by $MU_m = 1$ and the Marginal utility for consumption is given by $MU_c = 1/c$. For the remainder of this problem we will normalize the price of money to 1. Let p_c be the price of the consumption good.

11. Suppose that income is $I = 10$ and $p_c = 2$. What is the optimal consumption bundle (c^*, m^*) ?

- (a) (1/2, 9)
- (b) (5/2, 5)
- (c) (2, 6)
- (d) (5, 0)

12. Prof. Alejandro ran out of paper for printing this exam, so he asked Shynar for help. Shynar had to spend most of her weekly income on buying extra reams of paper so that the ECON101 students could have their exam. This left Shynar with an income $I = 1/2$. Suppose that the price of the consumption good is still $p_c = 2$. What is the optimal consumption bundle (c^*, m^*) ?

- (a) (0.193669, 0.112662)
- (b) (0.0261715, 0.447657)
- (c) (1/4, 0)
- (d) (0.241525, 0.0169492)

13. Suppose that Prof. Alejandro reimbursed Shynar and she is back to her weekly income of $I = 10$. Assume that the price of the consumption good is p_c . Shynar's Marshallian Demand system (c^M, m^M) is:

- (a) $(0.5/p_c, 9.5)$
- (b) $(1/p_c, 9)$
- (c) $(I/p_c, 0)$
- (d) $(0, 10)$

14. Keep assuming that income is $I = 10$ and that the price of p_c is left arbitrary. Shynar's *price consumption curve* is:
- (a) Horizontal and given by $m = 9$.
 - (b) Vertical and given by $c = 1/p_c$.
 - (c) $m = c$
 - (d) $m = c/p_c$
15. Suppose that Shynar is facing initially a price $p_c = 2$ for the consumption good, but that this price changes to $p_c = 1$.
- (a) The Income effect is 0 and the Substitution effect is 0.5.
 - (b) The Income effect is 0 and the Substitution effect is -0.5 .
 - (c) The Income effect is -0.25 and the Substitution effect is -0.25 .
 - (d) The Income effect is 0.5 and the Substitution effect is 0.
 - (e) The Income effect is 0.25 and the Substitution effect is 0.25.

Nozima wanted to make her large family happy and decided to cook something delicious for breakfast. She had 9 eggs in the fridge and a 5 kilogram pack of flour in the cupboard.

She knows that a mix of 140 grams of flour and 3 eggs yield enough batter for 6 crepes. Similarly, a mix of 140 grams of flour and 1 egg yields enough batter for 3 pancakes.

Thus, if c are crepes and p are pancakes, the recipes above can be represented by the following "utility functions":

$$c = 6 \min\{f/140, e/3\},$$

$$p = 3 \min\{f/140, e\},$$

where e are eggs and f is flour in grams.

16. With the amount of eggs and flour that she has in the pantry, how many pancakes can Nozima cook?
- (a) 37.5 pancakes.
 - (b) 27 pancakes.
 - (c) 107.1 pancakes.
 - (d) 9 pancakes.
17. With the amount of eggs and flour that she has in the pantry, how many crepes can Nozima cook?
- (a) 3 crepes.
 - (b) 225 crepes.
 - (c) 37.5 pancakes.
 - (d) 18 crepes.
18. Suppose that Nozima's family utility function for crepes and pancakes is given by $u(c, p) = \sqrt{c} + \sqrt{p}$. Suppose that they want to achieve a utility of $2\sqrt{3} + 3$ by consuming 12 crepes and 9 pancakes. Given the 9 eggs and 5000 grams of flour that Nozima has, how these ingredients should be allocated between crepes and pancakes?
- (a) 6 eggs and 3.3 kg of flour to crepes; 3 eggs and 1.6 kg of flour to pancakes.
 - (b) 6 eggs and 840 grams of flour to crepes; 3 eggs and 420 grams of flour to pancakes.
 - (c) 6 eggs and 0.084 kilograms of flour to crepes; 3 eggs and 0.042 kilograms of flour to pancakes.
 - (d) None of the above.

19. Suppose that Nozima discovers that her sister used 3 kilograms of flour to bake cake last week, so she only has 2 kilograms. Is the level of utility in the previous item for her family still attainable?
- (a) Yes it is, because she still can cook the same number of pancakes and crepes as before.
 - (b) No, because now she is not able to cook 12 crepes and the 9 pancakes required.
20. (Bonus question) Below you can find a picture of Panqueque.



From the nicknames below, guess what is Panqueque's correct nickname?

- (a) Gordo (Fatty in English, because he is a chonky-boy).
- (b) Mr. Lepu (because he has a furry tail just as the Looney Tunes character Pepe Lepu).
- (c) Bigotes de Foca (translated as Seal Whiskers because he has whiskers as large as a seal).
- (d) All of the above.